Task 1 - Foundation & Environment Setup

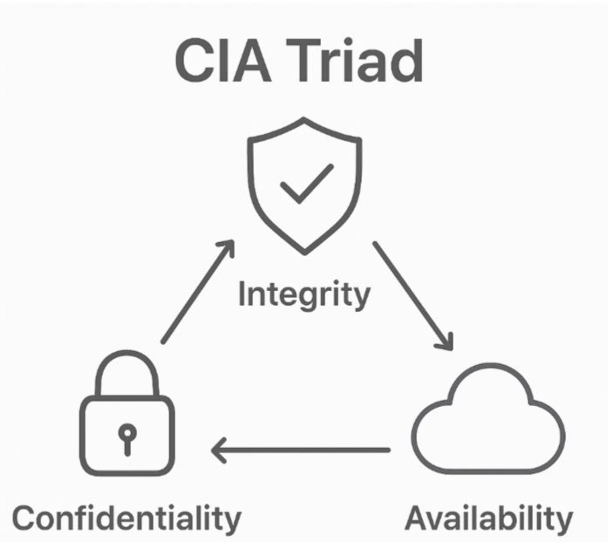
❖ Objective:

Build strong fundamentals in cybersecurity, networking, cryptography, and set up a professional hacking lab.

❖ Observation:

1. Cybersecurity Basics

≡ CIA Triad



Confidentiality is a set of rules that prevents sensitive information from being disclosed to unauthorized people, resources and processes. Methods to ensure confidentiality include data encryption, identity proofing and two factor authentications. **Integrity** ensures that system information or processes are protected from intentional or accidental modification. One way to ensure integrity is to use a hash function or checksum. **Availability** means that authorized users are able to access systems and data when and where needed and those that do not meet established conditions, are not. This can be achieved by maintaining equipment, performing hardware repairs, keeping operating systems and software up to date, and creating backups.

≡ Threat Types

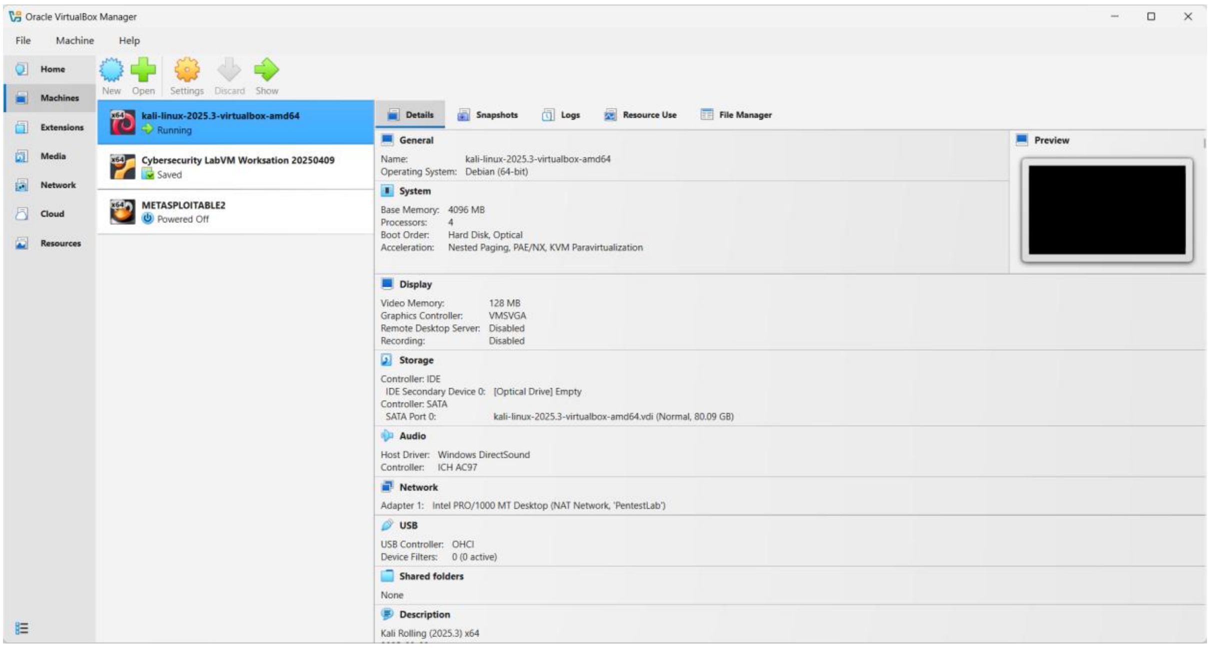
- Phishing: A social engineering attack where attackers impersonate a trusted entity (via email, message, or website) to trick victims into revealing sensitive information like passwords or financial data.
- Malware: Malicious software (viruses, worms, trojans, spyware, etc.) designed to infiltrate, damage, or gain unauthorized access to a computer system.
- DDoS (Distributed Denial of Service): An attack that overwhelms a target system, server, or network with a flood of traffic from multiple sources, making it unavailable to legitimate users.
- SQL Injection: A code injection technique where malicious SQL statements are inserted into input fields to manipulate or exploit a database, often to access, modify, or delete data.
- Brute Force: An attack method that systematically tries all possible combinations of passwords or encryption keys until the correct one is found.
- Ransomware: A type of malware that encrypts a victim's files or locks them out
- of their system, demanding payment (ransom) for restoring access.

≡ Attack Vectors

- Social Engineering: The psychological manipulation of individuals into divulging confidential information or performing actions that compromise security, exploiting human trust rather than technical vulnerabilities.
- Wireless Attacks: Attacks targeting vulnerabilities in wireless networks (Wi-Fi) to intercept data, gain unauthorized access, or disrupt network communication.
- Insider Threats: Security risks originating from individuals within an organization (employees, contractors, etc.) who misuse their authorized access to harm systems, steal data, or compromise security, whether intentionally or accidentally.

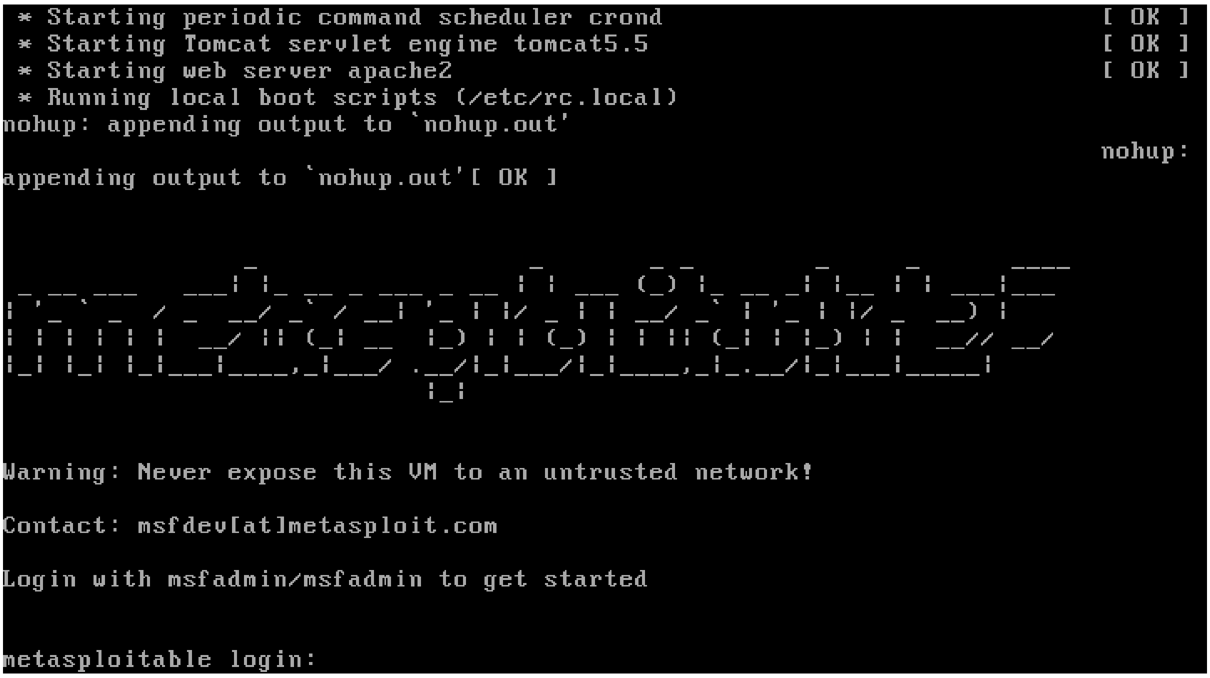
2. Lab Environment Setup

≡ Install VirtualBox

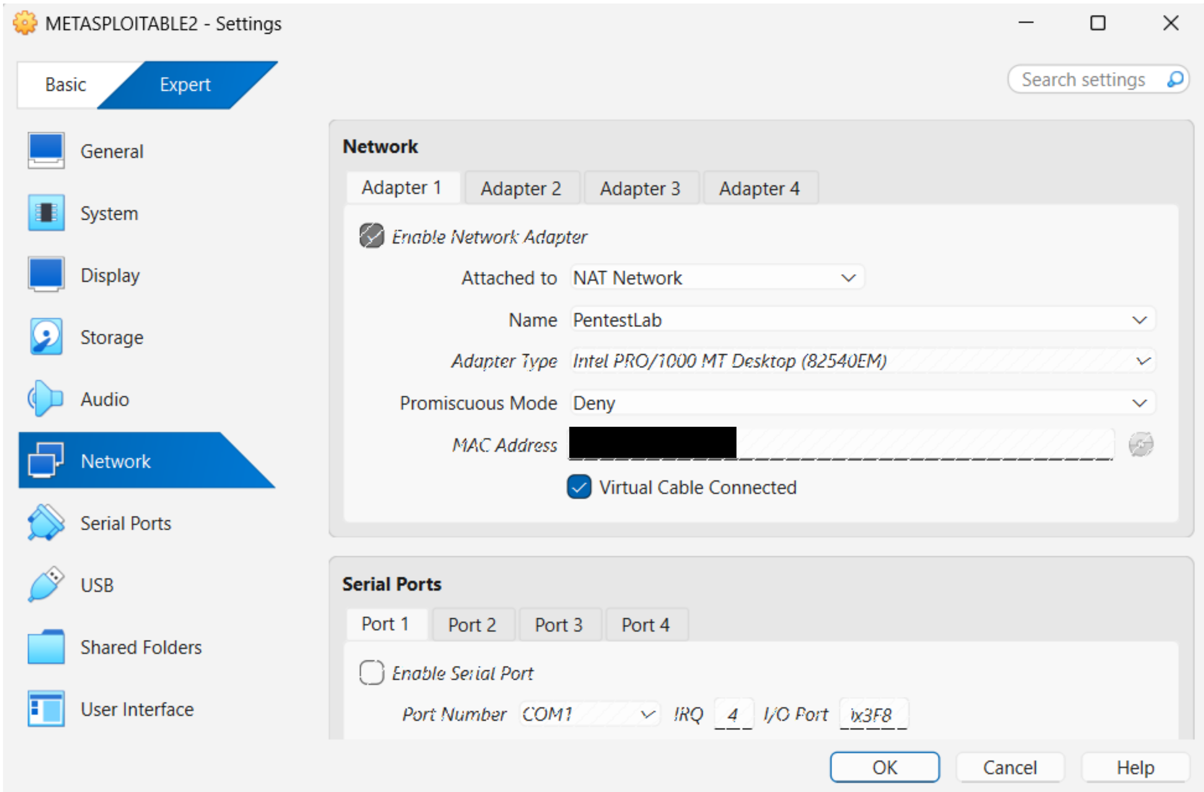


≡ Install Kali Linux as the attacker machine

☰ Install Metasploitable2 as target machine



☰ Configure a private lab network (Host-Only Adapter).



3. Linux Fundamentals

File System Navigation

```
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ pwd
/mnt/e/ApexPlanet Software/Task1
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ ls -la
total 0
drwxrwxrwx 1 student student 4096 Jul 26 19:05 .
drwxrwxrwx 1 student student 4096 Jul 26 19:06 ..
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ cd ..
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software$ cd Task1
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ cd ~
student@LAPTOP-6R9RA85I:~$
```

File & Directory Permissions

```
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ ls -l file.txt
-rwxrwxrwx 1 student student 0 Jul 26 19:10 file.txt
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ chmod 755 script.sh
```

Package Management

```
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [881 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1141 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [196 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.3 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1197 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [276 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [180 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1676 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [238 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.4 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [333 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1240 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1348 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [302 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [45.4 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.1 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5760 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [32.5 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.7 kB]
Get:26 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [281 kB]
Get:27 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [40.3 kB]
Get:28 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [10.4 kB]
Fetched 10.3 MB in 4s (2388 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
69 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Networking Commands

```
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ ping -c 4 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.

--- 10.0.2.15 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3081ms
```

```
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ traceroute 10.0.2.15
traceroute to 10.0.2.15 (10.0.2.15), 30 hops max, 60 byte packets
 1  LAPTOP-6R9RA85I.mshome.net (172.20.16.1)  4.133 ms  2.668 ms  2.625 ms
 2  RTK_GW.bbrouter (192.168.1.1)  11.143 ms  11.046 ms  10.972 ms
 3  10.230.192.1 (10.230.192.1)  11.686 ms  12.732 ms  12.662 ms
```

```
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether [redacted] brd ff:ff:ff:ff:ff:ff
    inet 172.20.26.71/20 brd 172.20.31.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe53:24ea/64 scope link
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether [redacted] brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
student@LAPTOP-6R9RA85I:/mnt/e/ApexPlanet Software/Task1$ ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 172.17.0.1  netmask 255.255.0.0  broadcast 172.17.255.255
        ether [redacted]  txqueuelen 0  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 2 overruns 0  carrier 0  collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.20.26.71  netmask 255.255.240.0  broadcast 172.20.31.255
        inet6 fe80::215:5dff:fe53:24ea  prefixlen 64  scopeid 0x20<link>
        ether [redacted]  txqueuelen 1000  (Ethernet)
        RX packets 17942  bytes 26412094 (26.4 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3814  bytes 299394 (299.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 74  bytes 9049 (9.0 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 74  bytes 9049 (9.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

4. Networking Basics

OSI Model Layers &Functions

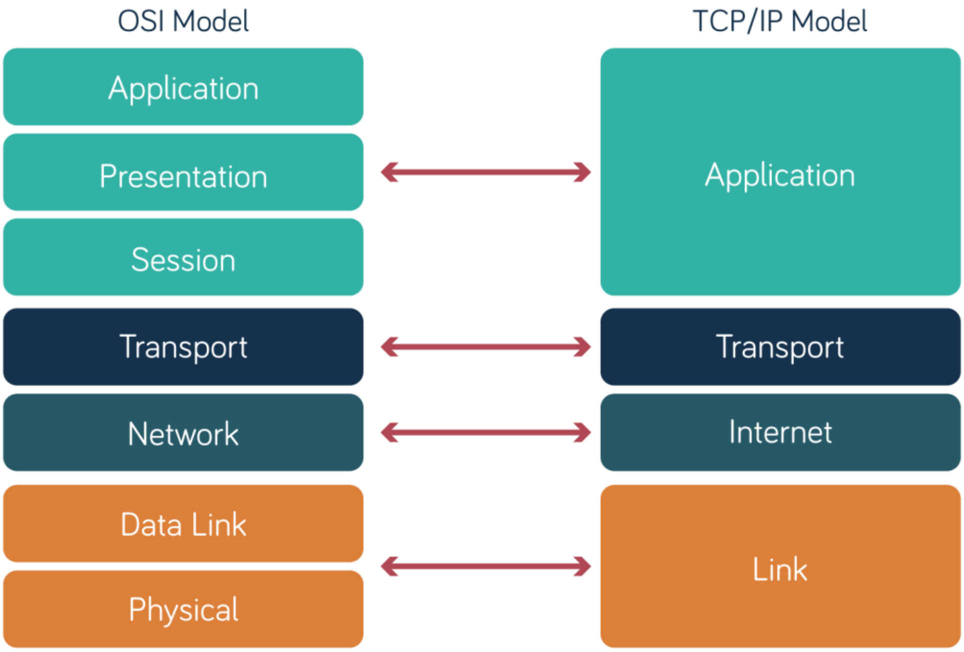
The OSI (Open Systems Interconnection) Model is a 7-layer conceptual framework that standardizes how data is transmitted across a network.

Layer	Name	Function
7	Application	Interface for end-user services (HTTP, FTP, SMTP, DNS)
6	Presentation	Data translation, encryption, compression (SSL/TLS, JPEG)
5	Session	Manages and terminates sessions between apps
4	Transport	End-to-end delivery, error control, flow control (TCP, UDP)
3	Network	Logical addressing and routing (IP, ICMP)
2	Data Link	Physical addressing, framing, error detection (switches, MAC)
1	Physical	Transmission of raw bits over physical medium (cables, signals)

TCP/IP Protocol Suite

A 4-layer practical model that governs real-world internet communication.

TCP/IP Layer	Corresponds to OSI	Key Protocols
Application	App + Presentation + Session	HTTP, FTP, DNS, SMTP
Transport	Transport	TCP, UDP
Internet	Network	IP, ICMP, ARP
Network Access	Data Link + Physical	Ethernet, Wi-Fi



DNS & HTTP/HTTPS Deep Dive

DNS (Domain Name System):

- ⇨ Translates human-readable domain names (e.g., google.com) into IP addresses.
- ⇨ Works hierarchically: Root → TLD (.com) → Authoritative Server.
- ⇨ Uses UDP port 53 (TCP for larger queries/zone transfers).

HTTP (HyperText Transfer Protocol):

- ⇨ Application-layer protocol for transferring web content.
- ⇨ Stateless — each request is independent.
- ⇨ Uses port 80. Data sent in plain text (insecure).

HTTPS (HTTP Secure):

- ⇨ HTTP layered over SSL/TLS encryption.
- ⇨ Uses port 443.
- ⇨ Ensures confidentiality, integrity, and authentication via certificates.

IP Addressing, Subnetting, and NAT

IP Addressing:

- ⇨ IPv4: 32-bit address (e.g., 192.168.1.1), divided into Classes A–E.
- ⇨ IPv6: 128-bit address, designed to solve IPv4 exhaustion.

Subnetting:

- ⇨ Dividing a large network into smaller sub-networks using a subnet mask (e.g., /24 = 255.255.255.0).
- ⇨ Improves security, reduces congestion, and organizes IP allocation.
- ⇨ CIDR (Classless Inter-Domain Routing) notation is used (e.g., 172.20.26.0/20).

NAT (Network Address Translation):

- ⇨ Translates private IP addresses (internal network) to a public IP and vice versa.
- ⇨ Types: Static NAT, Dynamic NAT, PAT (Port Address Translation).

5. Cryptography Basics

Symmetric vs Asymmetric Encryption

Symmetric Encryption	Asymmetric Encryption
Single shared key (same for encrypt & decrypt)	Key pair - Public key (encrypt) & Private key (decrypt)
Speed :- Fast	Speed :- Slower
Use case :- Bulk data encryption (files, disk)	Use case :- Secure key exchange, digital signatures
Example :- AES, DES, 3DES	Example :- RSA, ECC, DSA
Problem :- Yes, sharing the key securely is a challenge	Problem :- No, public key can be shared openly

Hashing (MD5, SHA256)

Hashing is a one-way function that converts input data into a fixed-size string (hash/digest), used for integrity verification, not encryption (cannot be reversed).

(a) MD5 (Message Digest 5):

- Produces a 128-bit hash.
- Considered insecure - vulnerable to collision attacks; not recommended for security purposes today.

(b) SHA-256 (Secure Hash Algorithm):

- Part of the SHA-2 family, produces a 256-bit hash.
- Currently considered secure and widely used (e.g., blockchain, password storage, TLS certificates).

Digital Certificates & SSL/TLS

(a) Digital Certificates:

- Electronic documents that bind a public key to an entity's identity, issued by a Certificate Authority (CA).
- Contains: Owner's public key, issuer info, validity period, digital signature.
- Follows the X.509 standard.

(b) SSL/TLS (Secure Sockets Layer / Transport Layer Security):

- Cryptographic protocols that secure communication over a network.
- TLS is the modern, more secure successor to SSL.
- Process (TLS Handshake):
  - I. Client Hello (proposes cipher suites)
  - II. Server Hello + sends its Digital Certificate
  - III. Key exchange (asymmetric encryption used briefly)
  - IV. Symmetric session key established for the rest of the session
  - V. Encrypted data transfer begins
- Ensures: Confidentiality (encryption), Integrity (hashing), Authentication (certificates).

Hands-on: Encrypt and Decrypt Messages Using OpenSSL

OpenSSL is a command-line toolkit for implementing SSL/TLS and general-purpose cryptography.

a) Symmetric Encryption (AES-256)

```
# Create message.txt
echo "Hello, this is a confidential message for my Cybersecurity Internship Task 1." > message.txt

# Encrypt
openssl enc -aes-256-cbc -salt -pbkdf2 -in message.txt -out message.enc -k MySecretPass123

# Decrypt
openssl enc -aes-256-cbc -d -pbkdf2 -in message.enc -out decrypted.txt -k MySecretPass123

# ─── View all file contents ───
echo "═══ message.txt ═══"
cat message.txt

echo "═══ message.enc (encrypted - unreadable binary) ═══"
cat message.enc

echo "═══ decrypted.txt ═══"
cat decrypted.txt
```

```
┌──(kali㉿kali)-[~/Documents/Task1_OpenSSL]
└─$ ./symmetric.sh
═══ message.txt ═══
Hello, this is a confidential message for my Cybersecurity Internship Task 1.
═══ message.enc (encrypted - unreadable binary) ═══
Salted__"l12��U̡ĦT�:�y�{�r�▒�9����S�2d�_�KM!To�� z_I� H#�ə�x"��S�2)k]��4�������
═══ decrypted.txt ═══
Hello, this is a confidential message for my Cybersecurity Internship Task 1.
```

b) Generate Hash (SHA256)

```
openssl dgst -sha256 message.txt
```

```
┌──(kali㉿kali)-[~/Documents/Task1_OpenSSL]
└─$ ./GenerateHash.sh
SHA2-256(message.txt)= 7137f9284e67dcebb1ce33fb92dbfd9f5622ea6009bfe5bbe1053d3ebe4cbbf3
```

c) Asymmetric Encryption (RSA)

```
echo "This is a Cybesecurity Lab Manual" > message.txt
openssl genrsa -out private.pem 2048
openssl rsa -in private.pem -pubout -out public.pem
openssl pkeyutl -encrypt -inkey public.pem -pubin -in message.txt -out message.enc
openssl pkeyutl -decrypt -inkey private.pem -in message.enc -out decrypted.txt

echo "═══ message.txt ═══"
cat message.txt
echo "═══ message.enc (encrypted - unreadable binary) ═══"
cat message.enc
echo "═══ decrypted.txt ═══"
cat decrypted.txt
```

```
┌──(kali㉿kali)-[~/Documents/Task1_OpenSSL]
└─$ ./Asymmetric.sh
writing RSA key
═══ message.txt ═══
This is a Cybesecurity Lab Manual
═══ message.enc (encrypted - unreadable binary) ═══

=δ�7r�4��ₐmpP�xP�,��>�W���D�▒��4��I/���S��0� [������g�A��E¶s�@h���QdiYP����\yZxZ ?δehW��S�D�i�l�-��X�T�t?Dδr��\�����s�
�ʀg���:a��œ��4ZK��E+ev)a+�~� <��S2]��p8�Ц��6ȷ���Vɉ<�vQ�t#������)Ou�����4:Ł�m=��¨�R


═══ decrypted.txt ═══
This is a Cybesecurity Lab Manual
```

d) Generate a Self-Signed SSL Certificate

```
openssl req -x509 -newkey rsa:2048 -keyout key.pem -out cert.pem -days 365 -nodes \
  -subj "/C=IN/ST=Gujarat/L=Ahmedabad/O=ApexPlanetSoftware/CN=localhost"
echo "═══ cert.pem (certificate) ═══"
cat cert.pem
```

```
═══ cert.pem (certificate) ═══
-----BEGIN CERTIFICATE-----
MIIDqTCCApGgAwIBAgIUcZzRfM8BdL22cEa5rTV2lQhoFoswDQYJKoZIhvcNAQEL
BQAwZDELMAkGA1UEBhMCSU4xEDAOBgNVBAgMB0d1amFyYXQxEjAQBgNVBAcMCUFo
bWVkYWJhZDEbMBkGA1UECgwSQXBleFBsYW5ldFNvZnR3YXJlMRIwEAYDVQQDDAls
b2NhbGhvc3QwHhcNMjYwODAxMTUyNjI1WhcNMjcwODAxMTUyNjI1WjBkMQswCQYD
[redacted]
MBaAFH/haYEIImYDlWc/YhO1DiZGuLhxMA8GA1UdEwEB/wQFMAMBAf8wDQYJKoZI
hvcNAQELBQADggEBAAzf5fmy6v6Lxh6Y6wnrz3oKtVx8WOqkKIJcLvoAF5Ojtr6Z
m+gAZSGcyC367BUya2eSG/Znj1PvWwWSO03jz9owbDh8W20CEEnRpYTbl69Cqleg
pQvEXWESjcVlX1uUXFfivKLZn/vQA/VczZAbtnZ0cVzrwzQ43tWRdooTw4dD7dLA
duhfrlox9OTA+vTINIZun2Na9Ep3VjzO9l7aJeugtrVziAuJWDx/WR4snxb1Sohf
16ijabw6H9LOqAuvDiz/xaag0BdnYMdrDzBKVneFW0rMgyaJ0ZEpuuldWVVSkHeT
hhurtLgaGX0dOrb97NByn0yHdwLZ8yxN2b6S3GI=
-----END CERTIFICATE-----
```

6. Tool Familiarization

- Wireshark (packet capture)

```
┌──(kali㉿kali)-[~/Documents/Task1_OpenSSL]
└─$ wireshark -v
Wireshark 4.4.7.
```

- Nmap (network scanning)

```
┌──(kali㉿kali)-[~/Documents/Task1_OpenSSL]
└─$ nmap -v
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-01 21:04 +0530
```

- Burp Suite (web proxy)

```
┌──(kali㉿kali)-[~/Documents/Task1_OpenSSL]
└─$ burpsuite --version
2026.3.2-46522 Burp Suite Community Edition
```

- Netcat (network debugging)

```
┌──(kali㉿kali)-[~/Documents/Task1_OpenSSL]
└─$ nc -h
[v1.10-50]
```

❖ Conclusion:

The lab environment was successfully configured. This environment is ready to support subsequent penetration-testing exercise.